LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	12
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Empennage structure, configuration of longitudinal stabilizer and concord and fore plane, layout of configuration of lateral stabilizer and control
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Empennage structure, configuration of longitudinal stabilizer and concord and fore plane, layout of configuration of lateral stabilizer and control
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of aircraft empennage diagrams; Safety: No physical handling of aircraft structures; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the empennage?
(b) Name the two main sections of the empennage.
(c) How does the empennage differ from the fuselage structure?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General empennage structure and elements • Longitudinal stabilizer configuration • Concord and fore plane layout • Lateral stabilizer configuration • Lateral stabilizer control mechanisms	L & Ds	PPT, IB	15 min
(b)	• Structural details of longitudinal stabilizer • Configuration of fore plane elements • Lateral stabilizer structural layout • Control surface integration details • Material usage in empennage sections	L & Ds	PPT, PDF	20 min
(c)	• Summary of empennage structural elements • Comparison of longitudinal and lateral layouts • Key structural element identification • Review of configuration standards • Clarification of specific structural details	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the structural difference between the longitudinal and lateral stabilizers.
(b) Identify the key structural elements of the fore plane configuration.
(c) Describe how the lateral stabilizer control is integrated into the empennage structure.

12. Home Assignment and Follow Up	Prepare a brief note on the structural elements of the empennage to be discussed in the next lesson regarding control surfaces and tabs.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Control surface and tabs, aerodynamics characteristics of trailing edge controls, types of tabs, all moving surface, mechanical and structural aspects of control surface design	Refer to the master syllabus reference list	LP 13

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min